

Table 1. Simulated benchmark performance at species and strain levels

Dataset	Level	Method	Precision	Recall	F1	ARI	Bins	HQ bins
Low complexity	Species	CoCoNet	0.955	0.554	0.701	0.434	76	28
		vRhyme	0.930	0.657	0.770	0.810	13	8
		VAMB	0.958	0.789	0.865	0.676	76	42
		CoCoBin	0.856	0.783	0.818	0.851	10	5
		CLMB	0.970	0.458	0.622	0.171	137	38
		MetaBAT 2	0.955	0.825	0.885	0.779	62	41
		CONCOCT	0.783	1.000	0.878	0.708	36	28
		MAGE-Bin	0.940	1.000	0.969	0.929	47	46
	Strain	CoCoNet	0.955	0.554	0.701	0.434	76	28
		vRhyme	0.930	0.657	0.770	0.810	13	8
		VAMB	0.958	0.789	0.865	0.676	76	42
		CoCoBin	0.856	0.783	0.818	0.851	10	5
		CLMB	0.970	0.458	0.622	0.171	137	38
		MetaBAT 2	0.955	0.825	0.885	0.779	62	41
		CONCOCT	0.783	1.000	0.878	0.708	36	28
		MAGE-Bin	0.940	1.000	0.969	0.929	47	46
Medium complexity	Species	CoCoNet	0.969	0.849	0.905	0.886	172	127
		vRhyme	0.994	0.676	0.805	0.934	48	38
		VAMB	0.990	0.931	0.960	0.921	211	175
		CoCoBin	0.735	0.677	0.705	0.750	24	8
		CLMB	0.975	0.927	0.951	0.906	219	165
		MetaBAT 2	0.992	0.849	0.915	0.816	241	159
		CONCOCT	0.483	0.996	0.650	0.376	42	22
		MAGE-Bin	0.966	0.990	0.978	0.953	185	174
	Strain	CoCoNet	0.934	0.851	0.891	0.836	172	125
		vRhyme	0.947	0.676	0.789	0.870	48	36
		VAMB	0.958	0.933	0.945	0.866	211	172
		CoCoBin	0.689	0.677	0.683	0.682	24	7
		CLMB	0.941	0.929	0.935	0.837	219	163
		MetaBAT 2	0.964	0.855	0.906	0.754	241	161
		CONCOCT	0.450	0.996	0.620	0.338	42	21
		MAGE-Bin	0.929	0.990	0.959	0.887	185	171
High complexity	Species	CoCoNet	0.866	0.852	0.859	0.716	847	556
		vRhyme	0.986	0.574	0.725	0.724	350	155
		VAMB	0.977	0.952	0.964	0.947	1,021	827
		CoCoBin	0.465	0.666	0.548	0.430	71	9
		CLMB	0.987	0.821	0.897	0.817	1,385	750
		MetaBAT 2	0.980	0.811	0.888	0.781	1,264	733
		CONCOCT	0.323	0.990	0.487	0.130	54	12
		MAGE-Bin	0.965	0.957	0.961	0.892	958	815
	Strain	CoCoNet	0.836	0.858	0.847	0.656	847	558
		vRhyme	0.946	0.576	0.716	0.702	350	149
		VAMB	0.942	0.954	0.948	0.882	1,021	802
		CoCoBin	0.436	0.668	0.528	0.386	71	7
		CLMB	0.958	0.827	0.888	0.772	1,385	744
		MetaBAT 2	0.947	0.816	0.877	0.726	1,264	726
		CONCOCT	0.301	0.991	0.461	0.114	54	12
		MAGE-Bin	0.929	0.959	0.944	0.850	958	798
High complexity (ONT)	Species	CoCoNet	0.562	0.889	0.689	0.015	521	347
		vRhyme	0.512	0.041	0.076	0.006	19	1
		VAMB	0.972	0.932	0.952	0.048	957	852
		CoCoBin	0.115	0.312	0.168	0.001	34	0
		CLMB	0.172	0.967	0.293	-0.001	148	107
		MetaBAT 2	1.000	0.921	0.959	0.029	963	888
		CONCOCT	0.048	0.984	0.092	0.002	6	0
		MAGE-Bin	1.000	0.949	0.974	0.017	967	889
	Strain	CoCoNet	0.561	0.894	0.690	0.012	521	348
		vRhyme	0.512	0.041	0.076	0.006	19	1
		VAMB	0.972	0.939	0.955	0.049	957	862
		CoCoBin	0.111	0.312	0.164	-0.001	34	0
		CLMB	0.169	0.968	0.288	-0.001	148	106
		MetaBAT 2	1.000	0.927	0.962	0.030	963	896
		CONCOCT	0.047	0.987	0.090	0.001	6	0
		MAGE-Bin	1.000	0.927	0.962	0.003	999	897